CHAPTER 6

Logarithm function

S44, S47, S49, S50

Before you start · exponential and inequities

Without a calculator, in six minutes. Respond briefly. Mobilize areas and diversion Composite. Then choose the targeted review indicated under the questions.

1. Resolve $e^x = 1$.

2. Give the sign of e^x.

3. Determine the area of $\frac{1}{x-3}$.

4. Derivating e^{2x}.

5. Resolve 2 $x - 1 > 0$.

6. Solve $e^x < e^2$.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Back to strictness exp growth and exp(0). **2.** Trace the exponential over the horizontal axis. **3.** Solve x-3=0 before taking the opposite. **4.** Name the inner function and its derivative; R36. **5.** Isolate x by two operations which keep order. **6.** Use strict exp growth to compare Exhibitors. Questions R serve as support for the teacher, even if their collective ritual is planned later.

Objectives

Use ln properties; solve equations and thresholds; drift and study functions Logarithmics.

Course notes

Definition and properties

The exponential is continuous, strictly increasing on $\mathbb{R}$, and his image is $]0, +\infty[$. For everything $x > 0$, there is a unique reality y such that $e^y = x$: we note it x.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x > 0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln \sqrt{a} = \tfrac{1}{2} \ln a.$$

Product proof. $e^{\ln a + \ln b} = e^{\ln a} e^{\ln b} = ab$; the uniqueness of the antecedent gives the formula. no similar formula for ln $(a + b)$.

Derivative, Variations and Limits

We admit it's driftable on $]0,+\infty[$. Derivating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; therefore $(\ln x)' = \frac{1}{x} > 0$. Ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty.$$

Its curve is symmetrical with that of the exponential in relation to $y = x$. The right $x = 0$ is a vertical asymptote. Its tangent in 1 gives ln $x \leq x - 1$ for $x > 0$.

Composition and Comparative Growth

If $u > 0$ is derivable, $(\ln u)' = \frac{u'}{u}$. For the integer $m \geq 1$,

$$\lim_{x\to+\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x\to 0^+} x^m \ln x = 0.$$

Demonstration in 0. Lay down $t = -\ln x$; when $x \to 0^+$, $t \to +\infty$ and $x \ln x = -te^{-t} \to 0$ per comparison of growth rates. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. In $+\infty$, pose $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

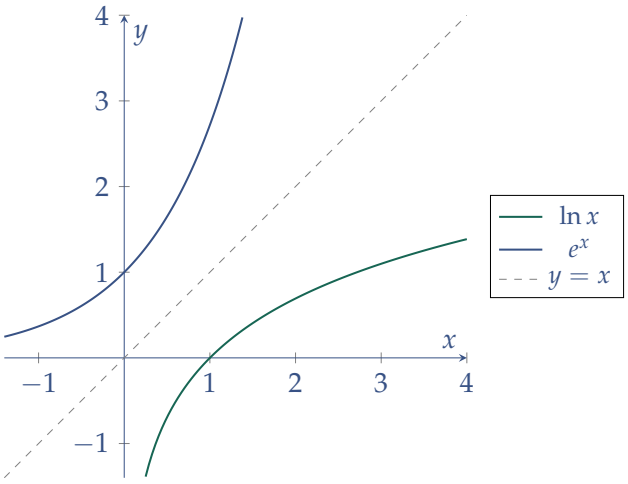
Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1) + \ln(x+1) =$ No 8 requires $x > 1$. Equation becomes $x^2 - 1 = 8$, so $x =$ Three alone.

Threshold. $0{,}8^n \leq 0.1$ is equivalent to $n \ln 0{,}8 \leq 0{,}1$, then $n \geq \frac{\ln 0{,}1}{\ln 0{,}8} \simeq 10{,}319$. The first integer is 11.

Example of study. For $f(x) = \frac{\ln x}{x}$, $f' = \frac{1-\ln x}{x^2}$; f growth on $]0, e]$, decreases on $[e, +\infty[$, and its maximum value $\frac{1}{e}$. Its limit in 0 $^+$ is $-\infty$, its limit in $+\infty$ is 0.



Use logarithm with its conditions

6.1 Inverse of exponential

First use: S44.

Definition · Neperian logarithm

For $x > 0$, $\ln x$ is the unique real number y such that $e^y = x$. Thus $e^{\ln x} = x$ for $x > 0$ and $\ln(e^y) = y$ for all real y. The logarithm transforms a positive value into a real exponent.

Example Solved · Domain before Transformation

To solve $\ln(x-1) = \ln(3-x)$, the two arguments must be positive, therefore $1 < x < 3$. The strict growth of ln then gives $x - 1 = 3 - x$, therefore $x = 2$, which belongs to the domain.

A1 · Immediate application Identify areas of $\ln(2-x)$ and $\ln(x^2-1)$. Solve $\ln(2x+1) = 0$ by specifying the field.

6.2 Products, quotients and powers

First use: S46.

Property · Algebraic Rules

For $a, b > 0$, $\ln(ab) = \ln a + \ln b$ and $\ln(a/b) = \ln a - \ln b$. For a relative integer m, $\ln(a^m) = m \ln a$. The conditions on the arguments remain to be checked after a transformation.

Demonstration. Lay down $u = \ln a$, $v = \ln b$. Then $ab = e^u e^v = e^{u+v}$, therefore $(ab) = u + v$. The identity of the quotient follows from $\ln(b \times b^{-1}) = \ln 1 = 0$, and that of the powers by repetition of the product then passage On the other hand. □

Example solved · An exact expression

$\ln 12 - \ln 3 = \ln 4 = 2 \ln 2$. By contrast, $\ln(1+1) = \ln 2 \neq \ln 1 + \ln 1 = 0$. Logarithm not distributive on the addition.

A2 · Two errors to explain Refuse " $\ln(a+b) = \ln a + \ln b$ Then explain why $\ln(x^2) = 2 \ln x$ is not an identity on $\mathbb{R} \setminus \{0\}$ and give the correct identity on this set.

6.3 Studying a logarithmic function

First use: S47.

Example Solved · Variation and Maximum

On $]0; +\infty[$, $f(x) = \frac{\ln x}{x}$ a for derivative $f'(x) = \frac{1 - \ln x}{x^2}$. It grows up to $x = e$, then decreases, and its maximum is $1/e$. Its sign depends on $\ln x$, so it is negative before 1 ; this does not prevent its growth on the $]0; 1[$.

A3 · Partial guidance Study the domain, derivative and sign of $g(x) = \ln(1 + x^2)$. Explain why the function is even and what it announces for variations.

6.4 The sign of $\ln q$ controls inequality

First use: S50.

Example Solved · A Lower Threshold

For $n \in \mathbb{N}$, $0{,}8^n < 0{,}1$ is equivalent to $n \ln 0{,}8 < \ln 0{,}1$. Like $\ln 0{,}8 < 0$, the inverse division Order: $n > \ln 0{,}1 / \ln 0{,}8 \simeq 10{,}32$. The first row is 11; $0{,}8^{10} \simeq 0{,}1074$ and $0{,}8^{11} \simeq 0{,}0859$ control the border. $q > 1$, $\ln q > 0$ and order is kept; for $q = 1$, we don't divides Not by ln $q = 0$.

A4 · Choosing common sense Determine the smallest integer $n \geq 0$ as $1{,}05^n \geq 2$, then as $0{,}9^n \leq 0.2$. Check the previous rank in each case.

6.5 Compare and change variable

First use: S49.

Ownership · Common limitations

$\ln x \to +\infty$ in $+\infty$ and $\ln x \to -\infty$ in 0^+. For the integer $m \geq 1$, $(\ln x)/x^m \to 0$ in $+\infty$; of plus $x \ln x \to 0$ in 0^+. The first comparative growth is derived from $t/e^{mt} \to 0$ with $t = \ln x$.

Solved Example · Adapting Comparative Growth

In $+\infty$, $\frac{2+\ln x}{x} = 2/x + \ln x/x \to 0$. In 0^+, $x^2 \ln x$ becomes $-te^{-2t}$ with t=-ln x; this expression tend towards 0.

A5 · Deferred application Calculate the limits of $\ln(2x)/x$ in $+\infty$ and $x \ln(3x)$ in 0^+. To detail a transformation using algebraic rules.

A6 · Synthesis problem Depending on the real a, determine the number of positive $\ln x = ax$. Use ln study x/xits limits and the position of the level a.

A first step: Distinguish $a < 0$, $a = 0$, $0 < a < 1/e$, $a = 1/e$ and $a > 1/e$. An unmet limit does not provide not an additional solution.

Wait a minute at the courthouse. Explain why "taking logarithm" is not allowed on a expression whose positivity has not been verified.

Review postponed · S58 and then S66 Identify a cooling constant, then find a antiderivative $1/x$ keeping the definition interval.

Research activity

Control a calculation error · 30 min.

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Note e_n error after n steps, for $n \in \mathbb{N}$.

1. Express e_n according to the n.

2. Using logarithm, determine the smallest integer n such that $e_n \leq 10^{-4}$.

3. Calculate, at the calculator, the error to the rank found and to the previous rank to check the minimality of the rank.

4. We now impose an error not more than 10^{-6}. Determine the new minimum number Does the division of tolerance by 100 multiply the number of steps per 100?

Progressive exercises

E1 · Direct application. After determining the set on which it is defined, resolve Equation $\ln(2x-1) = \ln(x+3)$.

E2 · Application and justification. Determining the real x for which $\ln x + \ln(x-2)$ is defined, then resolve $\ln x + \ln(x-2) = \ln 3$.

E3 · Method to be organised. It is considered $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. Clarify their set and to calculate their derivative.

E4 · Choice of strategy. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justifying growth results comparative employees.

E5 · Synthesis or demonstration. Using the study of a function, demonstrate that $\ln x \leq x-1$ for all real $x > 0$. The concavity of the logarithm shall not be used directly.

Work in the manual or annal

Barbazo, p. 196, Exercise 36. At home after S44: treating the three functions. before solving the inequities; justify the open bounds of the intervals obtained.

Duty at home

DM6 · Study of a logarithmic function · 55 min.

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{1+\ln x}{x}$. Its representative curve is noted $\mathcal{C}_f$.

1. Determining the limits of f at the boundaries of the domain, study his sign, then draw his picture changes.

2. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$.

3. Demonstrate that the equation $f(x) = \frac{1}{2}$ admits exactly two solutions. Give each one one justified framework using calculated values.

4. Using the maximum of f, demonstrate that $1 + \ln x \leq x$ for every $x > 0$. Specify the case equality.

QCM A · domain and properties

After S47; QCM B after S50.

For each question, only one proposal is accurate. consult the correction.

1. The domain of $\ln\ (2x-1)$ is:

A. $\mathbb{R}$ B. $[\frac{1}{2};+\infty[$

C. $]0;+\infty[$ D. $]\frac{1}{2};+\infty[$

2. For $a,b>0$, $\ln(ab)$ is valid for:

A. $\ln a+\ln b$ B. $\ln a\ln b$

C. $\ln(a+b)$ D. $a\ln b$

3. $\ln(e^{-3})$ is valid for:

A. 3 B. e^3

C. -3 D. $\ln 3$

4. The derivative of $\ln(1+x^2)$ is:

A. $\frac{1}{1+x^2}$ B. $\frac{2x}{1+x^2}$

C. $2\ln x$ D. $\frac{2}{x}$

5. The equation $\ln x=2$ a for solution:

A. $2e$ B. $\ln 2$

C. 2 D. e^2

QCM B · Threshold, Growth and Parameter

For each question, only one proposal is accurate.
consult the correction.

6. The smallest integer $n\geq 0$ as $(\frac{1}{2})^n<0.1$ is:

A. 3 B. 10

C. 4 D. 2

7. For $0<q<1$ and $0<a<1$, $q^n<a$ equals:

A. $n>\frac{\ln a}{\ln q}$ B. $n<\frac{\ln a}{\ln q}$

C. $n>\frac{\ln q}{\ln a}$ D. $n>a/q$

8. $\lim_{x\to+\infty}\frac{\ln x}{x}$ is valid for:

A. 1 B. 0

C. $+\infty$ D. $-\infty$

9. $\ln(x^2)=2\ln x$ is valid:

A. for all real x B. for everything $x\neq 0$

C. only for $x>1$ D. for everything $x>0$

10. The maximum of $g(x)=\frac{\ln x}{x}$, $x>0$, is:

A. 1 B. e

C. $1/e$ D. 0

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $\ln(1+1)=\ln 1+\ln 1$.

2. Divide inequality by ln 2 reverses its meaning.

3. $\ln x / x = 1$ has a positive solution.